UNIT - IV

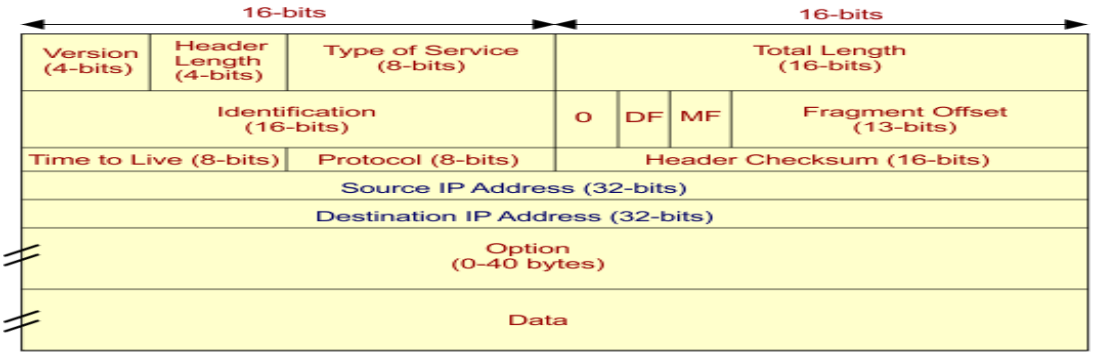
Syllabus: Network Layer Protocols: Network Layer-IPv4, IPv6, 6LoWPAN, 6TiSCH, ND, DHCP, ICMP, RPL, CORPL, CARP.

Network Layer Routing Protocols Internet Protocol Version 4 (IPv4)

IPv4 is 32-bit addressing scheme used as TCP/IP host addressing mechanism. IP addressing enables every host on the TCP/IP network to be uniquely identifiable. IPv4 is a connectionless protocol used for packet- switched networks. It operates on a best effort delivery model, in which neither delivery is guaranteed, nor is proper sequencing or avoidance of duplicate delivery assured. Internet Protocol Version 4 (IPv4) is the fourth revision of the Internet Protocol and a widely used protocol in data communication over different kinds of networks. IPv4 is a connectionless protocol used in packet-switched layer networks, such as Ethernet. It provides a logical connection between network devices by providing identification for each device. There are many ways to configure IPv4 with all kinds of devices – including manual and automatic configurations – depending on the network type. IPv4 is defined and specified in IETF publication RFC 791.IPv4 uses 32-bit addresses for Ethernet communication in five classes: A, B, C, D and E. Classes A, B and C have a different bit length for addressing the network host. Class D addresses are reserved for military purposes, while class E addresses are reserved for future use. IPv4 uses 32-bit (4 byte) addressing, which gives 232 addresses. IPv4 addresses are written in the dot-decimal notation, which comprises of four octets of the address expressed individually in decimal and separated by periods, for instance, 192.168.1.5.

IPv4 Datagram Header

Size of the header is 20 to 60 bytes.



IPV4 Datagram Header

VERSION: Version of the IP protocol (4 bits), which is 4 for IPv4
HLEN: IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.
Type of service: Low Delay, High Throughput, Reliability (8 bits)

Total Length: Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.
Identification: Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)
Flags: 3 flags of 1 bit each: reserved bit (must be zero), do not fragment flag, more fragments flag (same order)
Fragment Offset: Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes.
Time to live: Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.
Protocol: Name of the protocol to which the data is to be passed (8 bits)
Header Checksum: 16 bits header checksum for checking errors in the datagram header Source IP **address:** 32 bits IP address of the sender
Destination IP address: 32 bits IP address of the receiver
Option: Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not.
Due to the presence of options, the size of the datagram header can be of variable length (20 bytes to 60 bytes).
IPv4 provides hierarchical addressing scheme which enables it to divide the network into sub-networks, each with well-defined number of hosts. IP addresses are divided into many categories:
•Class A - it uses first octet for network addresses and last three octets for host addressing
•Class B - it uses first two octets for network addresses and last two for host addressing
•Class C - it uses first three octets for network addresses and last one for host addressing
•Class D - it provides flat IP addressing scheme in contrast to hierarchical structure for above three.
•Class E - It is used as experimental.
IPv4 also has well-defined address spaces to be used as private addresses (not routable on internet), and public addresses (provided by ISPs and are routable on internet).
Though IP is not reliable one; it provides 'Best-Effort-Delivery' mechanism

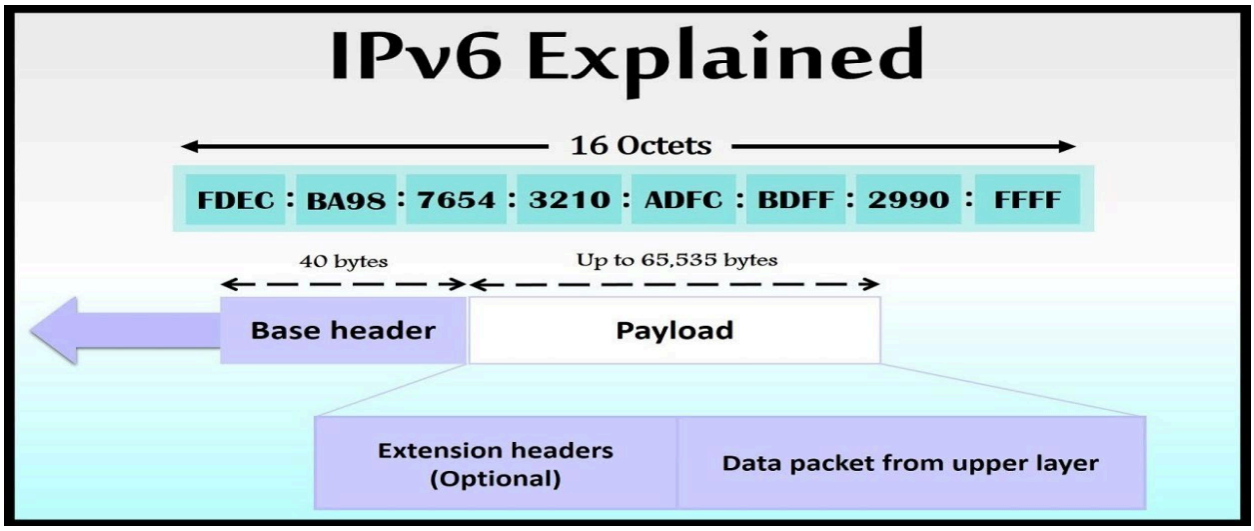
Internet Protocol Version 6 (IPv6)

IP v6 was developed by Internet Engineering Task Force (IETF) to deal with the problem of IP v4 exhaustion. IP v6 is a 128-bits address having an address space of 2^128, which is way bigger than IPv4. In IPv6 we use Colon-Hexa representation. There are 8 groups and each group represents 2 Bytes.

Exhaustion of IPv4 addresses gave birth to a next generation Internet Protocol version 6. IPv6 addresses its nodes with 128-bit wide address providing plenty of address space for future to be used on entire planet or beyond.

In IPv6 representation, have three addressing methods:

1) Unicast
2) Multicast
3) Anycast



1. Unicast Address –

Unicast Address identifies a single network interface. A packet sent to a unicast address is delivered to the interface identified by that address.

2. Multicast Address –

Multicast Address is used by multiple hosts, called as Group, acquires a multicast destination address. These hosts need not be geographically together. If any packet is sent to this multicast address, it will be distributed to all interfaces corresponding to that multicast address.

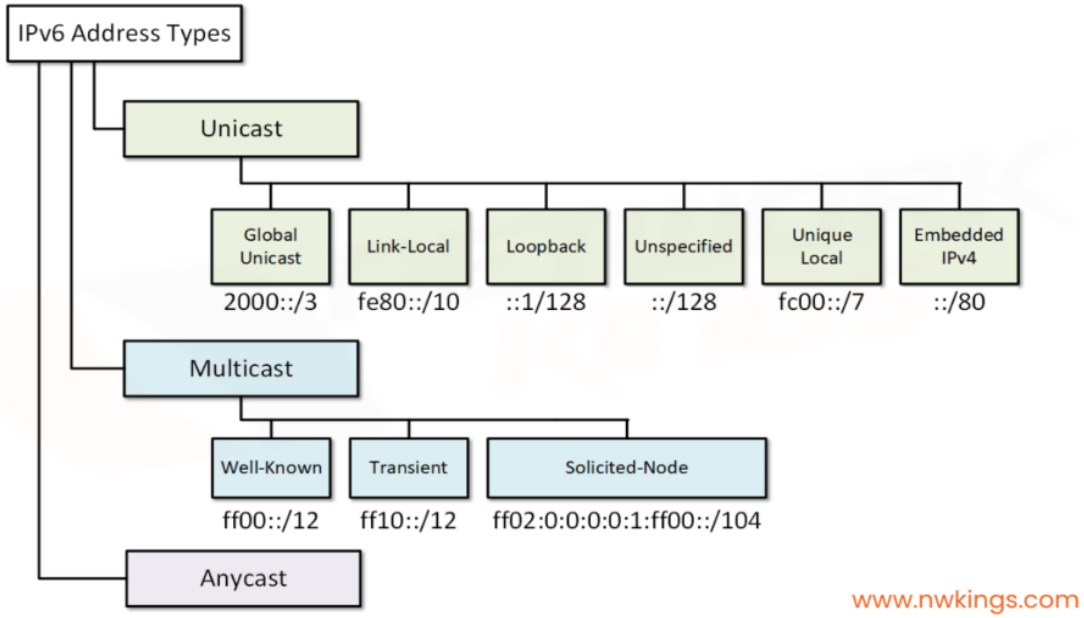
3. Anycast Address –

Anycast Address is assigned to a group of interfaces. Any packet sent to an anycast address will be delivered to only one member interface (mostly nearest host possible).

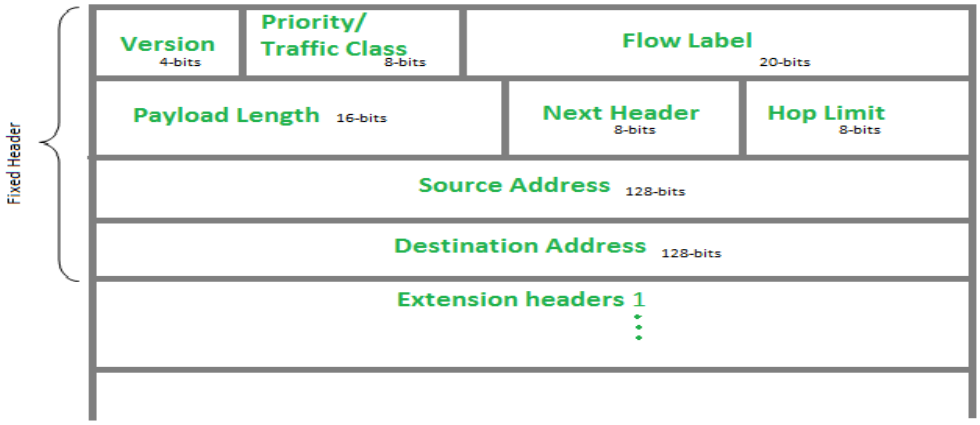
Note: Broadcast is not defined in IPv6.

Types of IPv6 address:

We have 128 bits in IPv6 address but by looking at the first few bits we can identify what type of address it is.



IP version 6 Header Format:



Version (4-bits): Indicates version of Internet Protocol which contains bit sequence 0110.
Traffic Class (8-bits): The Traffic Class field indicates class or priority of IPv6 packet which is similar to **Service Field in IPv4 packet.** It helps routers to handle the traffic based on the priority of the packet. If congestion occurs on the router then packets with the least priority will be discarded.
As of now, only 4-bits are being used (and the remaining bits are under research), in which 0 to 7 are assigned to Congestion controlled traffic and 8 to 15 are assigned to Uncontrolled traffic.

Priority assignment of Congestion controlled traffic:
Uncontrolled data traffic is mainly used for Audio/Video data. So we give higher priority to uncontrolled data traffic. The source node is allowed to set the priorities but on the way, routers can change it. Therefore, the destination should not expect the same priority which was set by the source node.

Priority	Meaning
0	No Specific traffic
1	Background data
2	Unattended data traffic
3	Reserved
4	Attended bulk data traffic
5	Reserved
6	Interactive traffic
7	Control traffic

Flow Label (20-bits): Flow Label field is used by a source to label the packets belonging to the same flow in order to request special handling by intermediate IPv6 routers, such as non-default quality of service or real-time service. In order to distinguish the flow, an intermediate router can use the source address, a destination address, and flow label of the packets. Between a source and destination, multiple flows may exist because many processes might be running at the same time. Routers or Host that does not support the functionality of flow label field and for default router handling, flow label field is set to 0. While setting up the flow label, the source is also supposed to specify the lifetime of the flow.

Payload Length (16-bits): It is a 16-bit (unsigned integer) field, indicates the total size of the payload which tells routers about the amount of information a particular packet contains in its payload. The payload Length field includes extension headers (if any) and an upper-layer packet. In case the length of the payload is greater than 65,535 bytes (payload up to 65,535 bytes can be indicated with 16-bits), then the payload length field will be set to 0 and the jumbo payload option is used in the Hop-by-Hop options extension header.

Next Header (8-bits): Next Header indicates the type of extension header(if present) immediately following the IPv6 header. Whereas In some cases it indicates the protocols contained within upper-layer packets, such as TCP, UDP.

Hop Limit (8-bits): Hop Limit field is the same as TTL in IPv4 packets. It indicates the maximum number of intermediate nodes IPv6 packet is allowed to travel. Its value gets decremented by one, by each node that forwards the packet and the packet is discarded if the value decrements to 0. This is used to discard the packets that are stuck in an infinite loop because of some routing error.

Source Address (128-bits): Source Address is the 128-bit IPv6 address of the original source of the packet.

Destination Address (128-bits): The destination Address field indicates the IPv6 address of the final destination (in most cases). All the intermediate nodes can use this information in order to correctly route the packet.

Extension Headers: In order to rectify the limitations of the IPv4 Option Field, Extension Headers are introduced in IP version 6. The extension header mechanism is a very important part of the IPv6 architecture. The next Header field of IPv6 fixed header points to the first Extension Header and this first extension header points to the second extension header and so on.



IPv6 has introduced Anycast addressing but has removed the concept of broadcasting. IPv6 enables devices to self-acquire an IPv6 address and communicate within that subnet. This auto-configuration removes the dependability of Dynamic Host Configuration Protocol (DHCP) servers. This way, even if the DHCP server on that subnet is down, the hosts can communicate with each other.

IPv6 provides new feature of IPv6 mobility. Mobile IPv6 equipped machines can roam around without the need of changing their IP addresses.

IPv6 is still in transition phase and is expected to replace IPv4 completely in coming years. At present, there are few networks which are running on IPv6. There are some transition mechanisms available for IPv6 enabled networks to speak and roam around different networks easily on IPv4. These are:

- Dual stack implementation
- Tunneling
- NAT-PT

Difference between IPv4 and IPv6:

IPv4	IPv6
IPv4 has a 32-bit address length	IPv6 has a 128-bit address length
It Supports Manual and DHCP address configuration	It supports Auto and renumbering address configuration
In IPv4 end to end, connection integrity is Unachievable	In IPv6 end-to-end, connection integrity is Achievable

IPv4	IPv6
It can generate 4.29×10^9 address space	The address space of IPv6 is quite large it can produce 3.4×10^{38} address space
The Security feature is dependent on the application	IPSEC is an inbuilt security feature in the IPv6 protocol
Address representation of IPv4 is in decimal	Address Representation of IPv6 is in hexadecimal
Fragmentation performed by Sender and forwarding routers	In IPv6 fragmentation is performed only by the sender
In IPv4 Packet flow identification is not available	In IPv6 packet flow identification are Available and uses the flow label field in the header
In IPv4 checksum field is available	In IPv6 checksum field is not available
It has a broadcast Message Transmission Scheme	In IPv6 multicast and anycast message transmission scheme is available
In IPv4 Encryption and Authentication facility not provided	In IPv6 Encryption and Authentication are provided
IPv4 has a header of 20-60 bytes.	IPv6 has a header of 40 bytes fixed
IPv4 can be converted to IPv6	Not all IPv6 can be converted to IPv4
IPv4 consists of 4 fields which are separated by addresses dot (.)	IPv6 consists of 8 fields, which are separated by a colon (:)
IPv4's IP addresses are divided into five different classes. Class A , Class B, Class C, Class Da , Class E.	IPv6 does not have any classes of the IP address.
IPv4 supports VLSM(Variable Length subnet mask).	IPv6 does not support VLSM.
Example of IPv4: 66.94.29.13	Example of IPv6: 2001:0000:3238:DFE1:0063:0000:0000:FEFB

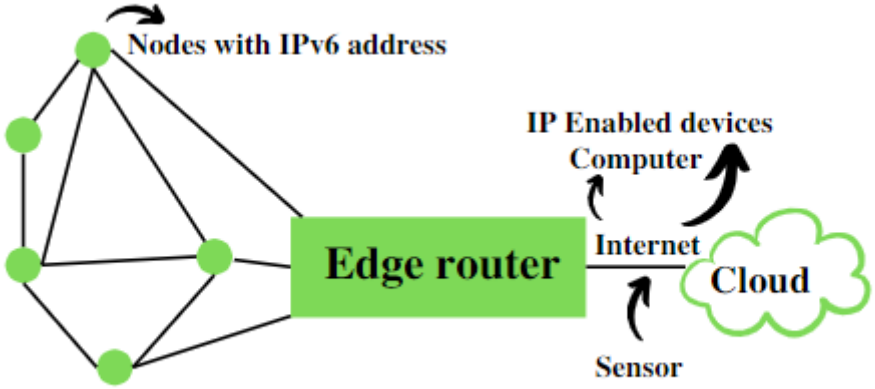
6LoWPAN:

6LoWPAN is an IPv6 protocol, and It's extended from is IPv6 over Low Power Personal Area Network. As the name itself explains the meaning of this protocol is that this protocol works on Wireless Personal Area Network i.e., WPAN.

WPAN is a Personal Area Network (PAN) where the interconnected devices are centered around a person's workspace and connected through a wireless medium. You can read more about WPAN at WPAN. 6LoWPAN allows communication using the IPv6 protocol. IPv6 is Internet Protocol Version 6 is a network layer protocol that allows communication to take place over the network. It is faster and more reliable and provides a large number of addresses.

6LoWPAN initially came into existence to overcome the conventional methodologies that were adapted to transmit information. But still, it is not so efficient as it only allows for the smaller devices with very limited processing ability to establish communication using one of the Internet Protocols, i.e., IPv6. It has very low cost, short-range, low memory usage, and low bit rate.

It comprises an Edge Router and Sensor Nodes. Even the smallest of the IoT devices can now be part of the network, and the information can be transmitted to the outside world as well. For example, LED Streetlights.



-It is a technology that makes the individual nodes IP enabled.
-6LoWPAN can interact with 802.15.4 devices and also other types of devices on an IP Network. For example, Wi-Fi.
-It uses AES 128 link layer security, which AES is a block cipher having key size of 128/192/256 bits and encrypts data in blocks of 128 bits each. This is defined in IEEE 802.15.4 and provides link authentication and encryption.

Basic Requirements of 6LoWPAN:

-The device should be having sleep mode in order to support the battery saving.
-Minimal memory requirement.
-Routing overhead should be lowered.

Features of 6LoWPAN:

-It is used with IEEE 802.15,.4 in the 2.4 GHz band.

-Outdoor range: ~200 m (maximum)

-Data rate: 200kbps (maximum)

-Maximum number of nodes: ~100

Advantages of 6LoWPAN:

-6LoWPAN is a mesh network that is robust, scalable, and can heal on its own.

-It delivers low-cost and secure communication in IoT devices.

-It uses IPv6 protocol and so it can be directly routed to cloud platforms.

-It offers one-to-many and many-to-one routing.

-In the network, leaf nodes can be in sleep mode for a longer duration of time.

Disadvantages of 6LoWPAN:

-It is comparatively less secure than Zigbee.

-It has lesser immunity to interference than that Wi-Fi and Bluetooth.

-Without the mesh topology, it supports a short range.

Applications of 6LoWPAN:

-It is a wireless sensor network.

-It is used in home-automation,

-It is used in smart agricultural techniques, and industrial monitoring.

-It is utilised to make IPv6 packet transmission on networks with constrained power and reliability resources possible.

Security and Interoperability with 6LoWPAN:

Security: 6LoWPAN security is ensured by the AES algorithm, which is a link layer security, and the transport layer security mechanisms are included as well.

Interoperability: 6LoWPAN is able to operate with other wireless devices as well which makes it interoperable in a network.

IPv6 over Low power Wireless Personal Area Network (6LoWPAN) is the first and most commonly used standard in this category. It efficiently encapsulates IPv6 long headers in IEEE802.15.4 small packets, which cannot exceed 128 bytes.

The specification supports different length addresses, low bandwidth, different topologies including star or mesh, power consumption, low cost, scalable networks, mobility, unreliability and long sleep time. The standard provides header compression to reduce transmission overhead, fragmentation to meet the 128-byte maximum frame length in IEEE802.15.4, and support of multi-hop delivery.

Frames in 6LoWPAN use four types of headers:

1) No 6loWPAN header (00),
2) Dispatch header (01),
3) Mesh header (10) and
4) Fragmentation header (11).

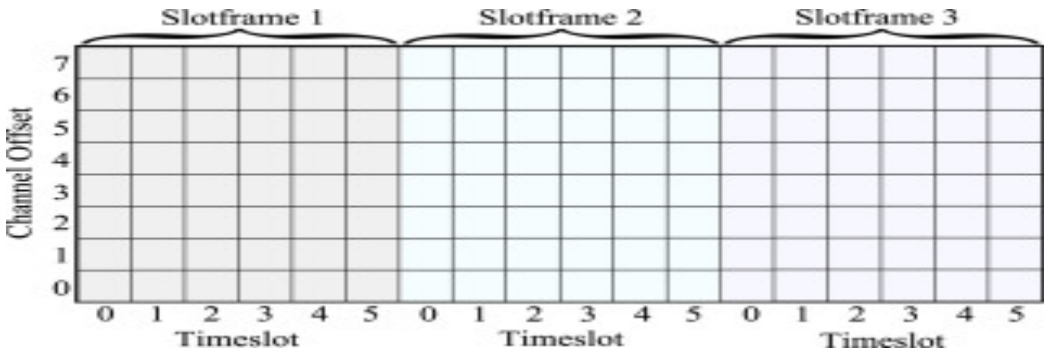
In No 6loWPAN header case, any frame that does not follow 6loWPAN specifications is discarded. Dispatch header is used for multicasting and IPv6 header compressions. Mesh headers are used for

broadcasting; while Fragmentation headers are used to break long IPv6 header to fit into fragments of maximum 128-byte length.

6 TiSCH:

The 6TiSCH working group aims to standardize IEEE802.15.4 Time-Slotted Channel Hopping (TSCH) with IPv6. Many of the problems related to bringing IPv6 to low capacity devices have been improved, such as 6LoWPAN Neighbor Discovery, IP-in-IP encapsulation, and header compression. In the 6TiSCH architecture, IEEE802.15.4 TSCH link layer is the root of the IETF 6TiSCH protocol suite. Some mechanisms are provided by this set of protocols, such as scheduling related policies, secure join procedures, minimal bootstrapping, and multi-hop communication through a mesh topology. Moreover, various network functionalities had been implemented, such as stateless and contextual header compression, fragmentation, neighbor discovery, and Routing Protocol for Low Power and Lossy Networks (RPL). RPL uses an objective function to optimize the multipoint-to-point routes, optimizing the communication between nodes and the root of the network, and point-to-multipoint so that the root is capable of reaching each node. In the application layer, it provides CoAP and OSCORE protocols to allow low-overhead secure RESTful interaction.

With such a robust suite constructed around IETF 6TiSCH, it allows the industrial performance of IEEE802.15.4 TSCH to use the IPv6 in low-cost devices. The TSCH protocol is based on the time-division multiple access (TDMA) with channel hopping for robustness against interference and multipath fading. Each timeslot must be wide enough to accommodate the frame followed by an acknowledgment, and a group of timeslots, called slotframe, repeat in time. Such behavior is shown in Fig. 1, where each slotframe has timeslots and there are channels. Note that, since each timeslot has available channels, there might be up to non-colliding communications per timeslot. In order to perform a communication between two nodes, the TSCH network must assign them a link (represent by each cell from table in Fig. 1), which means to reserve a channel offset and a timeslot so that the communication may occur.



6TiSCH working group in IETF is developing standards to allow IPv6 to pass through Time-Slotted Channel Hopping (TSCH) mode of IEEE 802.15.4e datalinks. It defines a Channel Distribution usage matrix consisting of available frequencies in columns and time-slots available for network scheduling operations in rows. This matrix is portioned into chucks where each chunk contains time and frequencies and is globally known to all nodes in the network. The nodes within the same interference domain negotiate their scheduling so that each node gets to transmit in a chunk within its interference

domain. Scheduling becomes an optimization problem where time slots are assigned to a group of neighboring nodes sharing the same application. The standard does not specify how the scheduling can be done and leaves that to be an application specific problem in order to allow for maximum flexibility for different IoT applications. The scheduling can be centralized or distributed depending on application or the topology used in the MAC layer.

6TiSCH, which stands for IPv6 over the TSCH mode of IEEE 802.15.4e, is a networking architecture designed for low-power and lossy networks (LLNs) in the Internet of Things (IoT) context. It is a standard developed by the Internet Engineering Task Force (IETF) to provide reliable and efficient communication in constrained IoT environments.

Here's a brief overview of the key concepts and components of 6TiSCH:

IPv6: 6TiSCH is built on the IPv6 protocol, which allows for the unique addressing and communication of devices in an IoT network. IPv6 provides a large address space, enabling the connection of a vast number of IoT devices.

Time-Slotted Channel Hopping (TSCH): 6TiSCH utilizes the TSCH mode of the IEEE 802.15.4e standard, which is a low-power wireless communication technology. TSCH divides time into timeslots and assigns specific channels for communication, allowing devices to schedule transmissions and reduce collisions.

Synchronization: In 6TiSCH, devices synchronize their communication schedules to avoid interference and collisions. The Time-Slotted Channel Hopping Synchronization Function (TSCH-SF) ensures that devices are aligned in time and channel usage.

Scheduled Communication: Devices in a 6TiSCH network communicate based on a predefined schedule, known as the schedule function (SF). This schedule specifies when and where each device can transmit and receive data, ensuring deterministic and reliable communication.

Routing: 6TiSCH employs a routing protocol called RPL (IPv6 Routing Protocol for Low-Power and Lossy Networks) for efficient routing within the network. RPL is designed to optimize energy consumption, adapt to changing network conditions, and support various topologies common in IoT deployments.

Security: 6TiSCH incorporates security mechanisms to protect IoT communication. It relies on existing IPv6 security extensions and can leverage additional security protocols such as DTLS (Datagram Transport Layer Security) for end-to-end encryption and authentication.

Overall, 6TiSCH provides a standardized approach to building low-power, highly reliable, and scalable IoT networks. By combining IPv6, TSCH, synchronization, scheduling, routing, and security, it enables efficient communication and interoperability among IoT devices in various application domains, such as industrial automation, smart cities, and smart homes.

6Lo:

Developed by IETF group, IPv6 over Networks of Resource-constrained Nodes (6Lo) is IPv6 based network access protocol for data links that were left excluded by 6LoWPAN and 6TiSCH. For example, IPv6 standardization over BLE, NFC, IEEE 802.11ah and many other MAC layer protocols will be included in this standard.

IPv6 over Bluetooth Low Energy –

This is an IPv6 adaption layer standard for Bluetooth 4.0 MAC layer protocol. The standard excludes the fragmentation feature of 6LoWPAN as there is already segmentation and reassembly feature of Logical Link Control and Adaptation Protocol (L2CAP) packets within BLE stack.

IPv6 over G.9959

This is an IPv6 addressing standard for G.9959 MAC layer protocol. G.9959 is a network access protocol stack designed for wireless networking of low power devices in a Personal Area Network (PAN).

Neighbor Discovery (ND):

Neighbor Discovery (ND) refers to a key mechanism in the IPv6 protocol suite that allows devices to discover and communicate with neighboring devices on the same network segment. It plays a crucial role in enabling device communication, address resolution, and neighbor reachability detection in IPv6-based IoT networks.

Here are the main components and functionalities of Neighbor Discovery in IoT:

Neighbor Discovery Protocol (NDP): NDP is a core component of IPv6 and encompasses several message types and functions for neighbor discovery. It operates at the network layer and is responsible for tasks such as neighbor solicitation, neighbor advertisement, and router discovery.

Address Auto configuration: NDP enables devices to automatically configure their IPv6 addresses based on the network prefix received from routers. With address autoconfiguration, IoT devices can assign themselves globally unique IPv6 addresses without the need for manual configuration or a centralized address management system.

Neighbor Solicitation (NS): NS messages are used by an IPv6 device to determine the link-layer address of a neighboring device. When an IoT device wants to communicate with a specific device, it sends an NS message to the solicited node multicast address of the target device, requesting its link-layer address.

Neighbor Advertisement (NA): NA messages are the responses to NS messages. When an IoT device receives an NS message, it replies with an NA message containing its link-layer address. This allows devices to establish neighbor relationships and maintain a mapping of IPv6 addresses to link-layer addresses.

Duplicate Address Detection (DAD): NDP includes a mechanism called Duplicate Address Detection, which ensures that the IPv6 address being assigned to a device is unique within the network. It helps prevent address conflicts and ensures proper address assignment in IPv6-based IoT deployments.

Router Discovery: NDP facilitates the discovery of routers in the network. IoT devices use router discovery to determine the presence of routers and obtain routing information, such as the default gateway and prefix information, which is essential for proper network connectivity.

Neighbor Discovery in IoT networks enables devices to discover and communicate with each other, resolve IPv6 addresses to link-layer addresses, and obtain essential network information from routers. It plays a vital role in facilitating device-to-device communication, enabling efficient routing, and ensuring the overall connectivity and functioning of IPv6-based IoT deployments.

Dynamic Host Configuration Protocol (DHCP):

Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to dynamically assign an IP address to nay device, or node, on a network so they can communicate using IP (Internet Protocol). DHCP automates and centrally manages these configurations. There is no need to manually assign IP addresses to new devices. Therefore, there is no requirement for any user configuration to connect to a DHCP based network.

DHCP can be implemented on local networks as well as large enterprise networks. DHCP is the default protocol used by the most routers and networking equipment. DHCP is also called RFC (Request for comments) 2131.

DHCP does the following:

-DHCP manages the provision of all the nodes or devices added or dropped from the network.
-DHCP maintains the unique IP address of the host using a DHCP server.
-It sends a request to the DHCP server whenever a client/node/device, which is configured to work with DHCP, connects to a network. The server acknowledges by providing an IP address to the client/node/device.
-DHCP is also used to configure the proper subnet mask, default gateway and DNS server information on the node or device.

There are many versions of DCHP are available for use in IPV4 (Internet Protocol Version 4) and IPV6 (Internet Protocol Version 6).
DHCP runs at the application layer of the TCP/IP protocol stack to dynamically assign IP addresses to DHCP clients/nodes and to allocate TCP/IP configuration information to the DHCP clients. Information includes subnet mask information, default gateway, IP addresses and domain name system addresses.
DHCP is based on client-server protocol in which servers manage a pool of unique IP addresses, as well as information about client configuration parameters, and assign addresses out of those address pools.

The DHCP lease process works as follows:

First of all, a client (network device) must be connected to the internet.
DHCP clients request an IP address. Typically, client broadcasts a query for this information.
DHCP server responds to the client request by providing IP server address and other configuration information. This configuration information also includes time period, called a lease, for which the allocation is valid.

When refreshing an assignment, a DHCP clients request the same parameters, but the DHCP server may assign a new IP address. This is based on the policies set by the administrator.

Components of DHCP

When working with DHCP, it is important to understand all of the components. Following are the list of components:

DHCP Server: DHCP server is a networked device running the DCHP service that holds IP addresses and related configuration information. This is typically a server or a router but could be anything that acts as a host, such as an SD-WAN appliance.

DHCP client: DHCP client is the endpoint that receives configuration information from a DHCP server. This can be any device like computer, laptop, IoT endpoint or anything else that requires connectivity to the network. Most of the devices are configured to receive DHCP information by default.

IP address pool: IP address pool is the range of addresses that are available to DHCP clients. IP addresses are typically handed out sequentially from lowest to the highest.

Subnet: Subnet is the partitioned segments of the IP networks. Subnet is used to keep networks manageable.

Lease: Lease is the length of time for which a DHCP client holds the IP address information. When a lease expires, the client has to renew it.

DHCP relay: A host or router that listens for client messages being broadcast on that network and then forwards them to a configured server. The server then sends responses back to the relay agent that passes them along to the client. DHCP relay can be used to centralize DHCP servers instead of having a server on each subnet.

Benefits of DHCP

There are following benefits of DHCP:

1) **Centralized administration of IP configuration:** DHCP IP configuration information can be stored in a single location and enables that administrator to centrally manage all IP address configuration information.
2) **Dynamic host configuration:** DHCP automates the host configuration process and eliminates the need to manually configure individual host. When TCP/IP (Transmission control protocol/Internet protocol) is first deployed or when IP infrastructure changes are required.

3) **Seamless IP host configuration:** The use of DHCP ensures that DHCP clients get accurate and timely IP configuration IP configuration parameter such as IP address, subnet mask, default gateway, IP address of DND server and so on without user intervention.

4) **Flexibility and scalability:** Using DHCP gives the administrator increased flexibility, allowing the administrator to move easily change IP configuration when the infrastructure changes.

Advantages

-Easy use and configuration, as the network parameters have to be entered only once.

-Furthermore, the existing IP addresses can be used optimally.

-Because of the frequently changed locations, it is almost impossible that mobile devices to configure themselves constantly. Here, the automatic DHCP system offers the advantage of uncomplicated and simple applicability.

Disadvantages

-The problem with DHCP is that clients accept any server. Accordingly, when another server is in the vicinity, the client may connect with this server, and this server may possibly send invalid data to the client.

-Security of MAC address is also not provided.

IP addresses may be static or dynamic. Static means that the address assigned to a machine generally does not change. Dynamic means that an IP address is assigned to a computer on demand, for a fixed lease period. The computer may be assigned a different address each time it demands one. Dynamic addresses are acceptable for a machine running client software, since the way things are organized is that the client initiates communications with a server, and includes its "return address" (the source IP address) in every packet sent to the server. To communicate to a server, it is necessary to find out the numeric IP address of the server before the client can communicate to it. That is often accomplished through the Domain Name System (DNS), essentially tables where the IP address of a server can be looked up. To avoid having to frequently update those tables, servers are generally assigned static addresses.

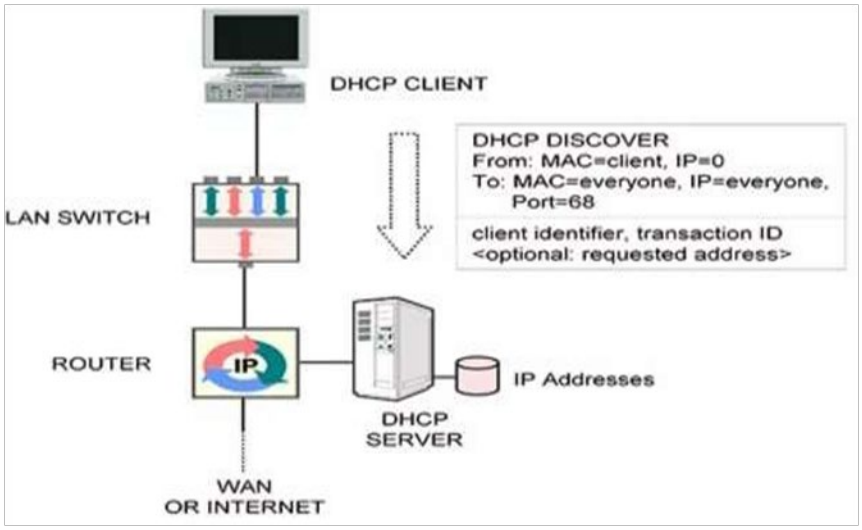


Fig: DHCP Discover message

Addresses are assigned to a computer using the Dynamic Host Configuration Protocol (DHCP). The system administrator provisions a DHCP server, configured to assign IP addresses within a defined range to clients. Computers are loaded with DHCP client software. Communications between the DHCP client and server are effectively application-layer messages, coded into ASCII and carried in UDP protocol data units, which are carried in IP packets, which are carried in MAC frames. The desired recipient of the messages is indicated as being the DHCP on a machine by populating in the UDP header destination port = 67 for messages to the server and destination port = 68 for messages to the client. The messages are "broadcast", which means that the destination IP address is all 1s and destination MAC address is all 1s.

The actual addresses are used for source MAC and IP addresses, except that the client uses "0" as its IP address, since of course the whole point of the exercise is to get an IP address. Each computer will run a DHCP client when it starts, generating a DHCP Discover message as illustrated above.

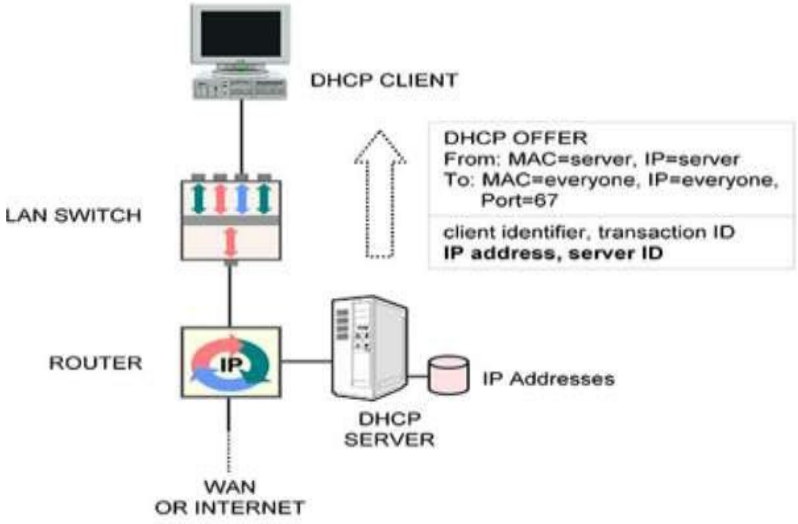


Fig: DHCP offer message

Any DHCP server that receives it, and there may be more than one, will respond with a DHCP Offer message, with an offered IP address and a lease time:

The client will answer with a DHCP Request message to confirm its selection of an offered address:

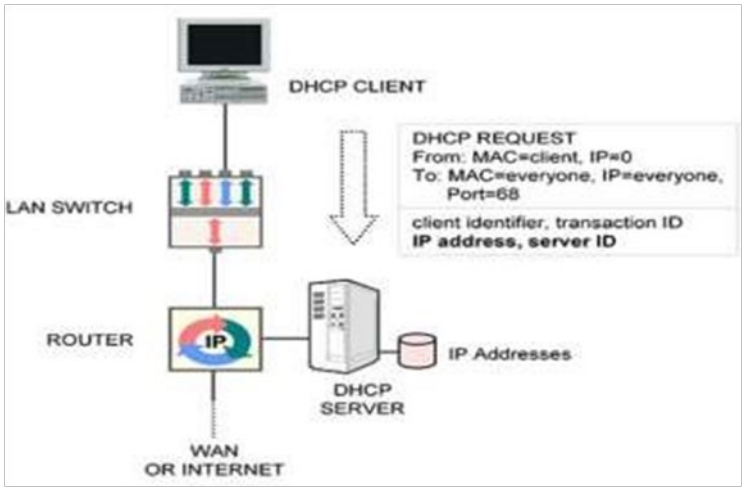


Fig: DHCP Request message

Then the server will complete the cycle with a DHCP ACK that usually includes other configuration information such as the IP address of the default gateway (CE router), the IP address of one or more DNS servers and the subnet mask, which indicates what bits in the address are the host or machine ID:

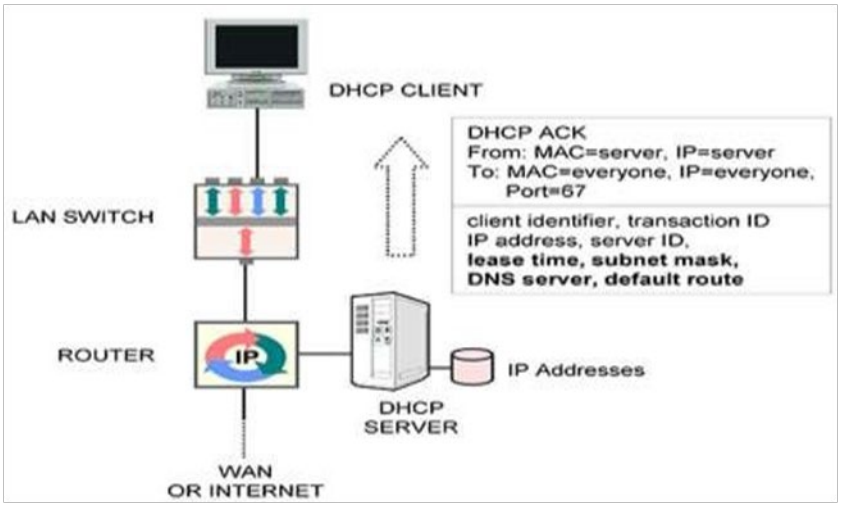


Fig: DHCP Acknowledge message

ICMP Protocol

The ICMP stands for Internet Control Message Protocol. It is a network layer protocol. It is used for error handling in the network layer, and it is primarily used on network devices such as routers. As different types of errors can exist in the network layer, so ICMP can be used to report these errors and to debug those errors.

For example, some sender wants to send the message to some destination, but the router couldn't send the message to the destination. In this case, the router sends the message to the sender that I could not send the message to that destination.

The IP protocol does not have any error-reporting or error-correcting mechanism, so it uses a message to convey the information. For example, if someone sends the message to the destination, the message is somehow stolen between the sender and the destination. If no one reports the error, then the sender might think that the message has reached the destination. If someone in-between reports the error, then the sender will resend the message very quickly.

Position of ICMP in the network layer

The ICMP resides in the IP layer, as shown in the below diagram.



Messages

The ICMP messages are usually divided into two categories:

ICMP messages

Category	Type	Message
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply

Error-reporting messages

The error-reporting message means that the router encounters a problem when it processes an IP packet then it reports a message.

Query messages

The query messages are those messages that help the host to get the specific information of another host. For example, suppose there are a client and a server, and the client wants to know whether the server is live or not, then it sends the ICMP message to the server.

ICMP Message Format

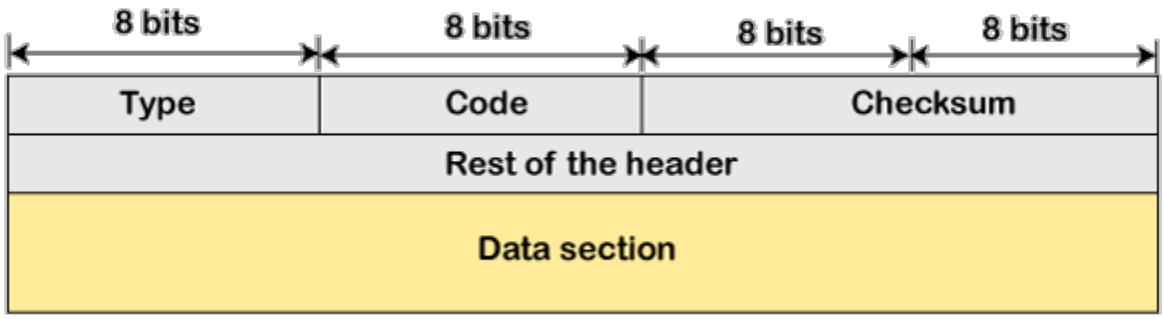
The message format has two things; one is a category that tells us which type of message it is. If the message is of error type, the error message contains the type and the code. The type defines the type of message while the code defines the subtype of the message.

The ICMP message contains the following fields:

Type: It is an 8-bit field. It defines the ICMP message type. The values range from 0 to 127 are defined for ICMPv6, and the values from 128 to 255 are the informational messages.

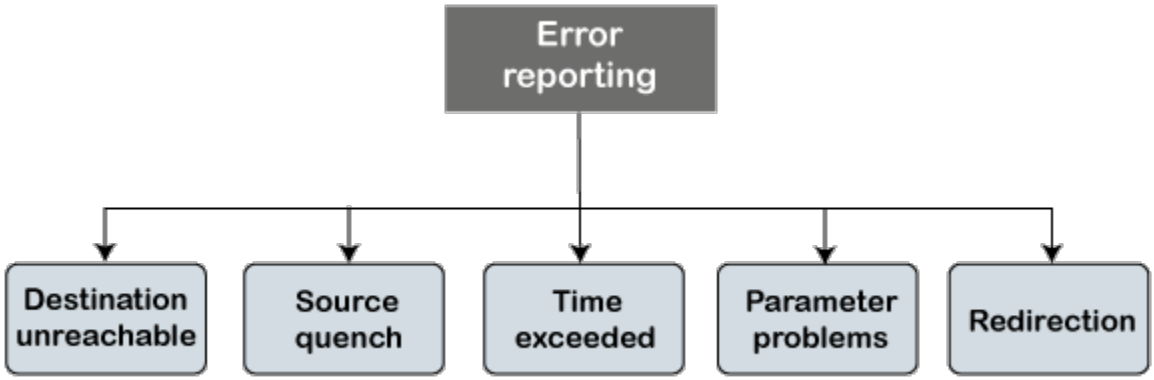
Code: It is an 8-bit field that defines the subtype of the ICMP message

Checksum: It is a 16-bit field to detect whether the error exists in the message or not.



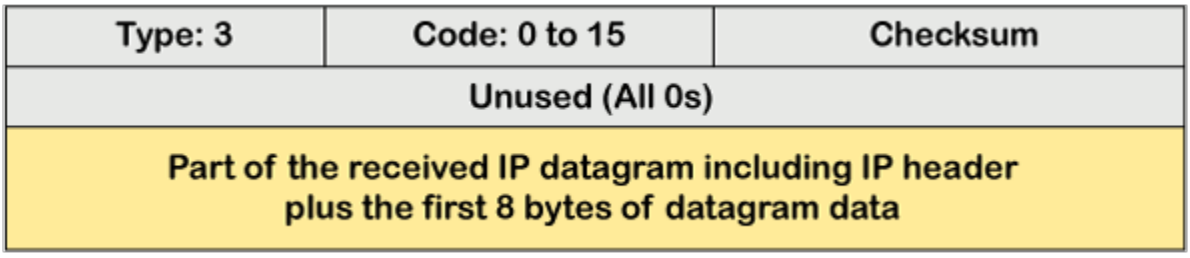
Types of Error reporting messages

The error reporting messages are broadly classified into the following categories:



Destination unreachable

The destination unreachable error occurs when the packet does not reach the destination. Suppose the sender sends the message, but the message does not reach the destination, then the intermediate router reports to the sender that the destination is unreachable.



The above diagram shows the message format of the destination unreachable message. In the message format:

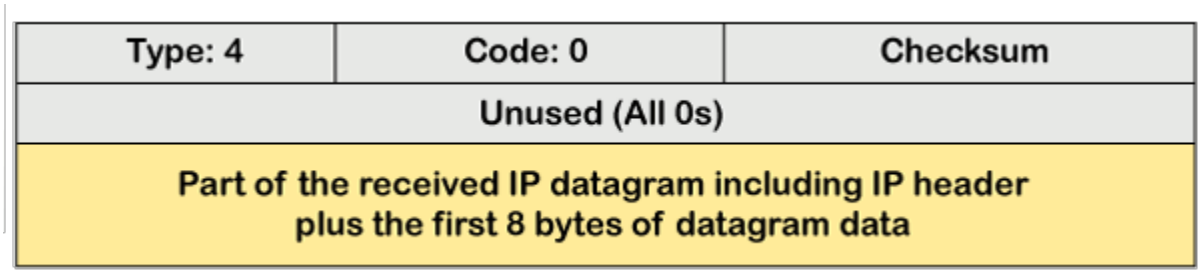
Type: It defines the type of message. The number 3 specifies that the destination is unreachable.

Code (0 to 15): It is a 4-bit number which identifies whether the message comes from some intermediate router or the destination itself.

Sometimes the destination does not want to process the request, so it sends the destination unreachable message to the source. A router does not detect all the problems that prevent the delivery of a packet.

Source quench

There is no flow control or congestion control mechanism in the network layer or the IP protocol. The sender is concerned with only sending the packets, and the sender does not think whether the receiver is ready to receive those packets or is there any congestion occurs in the network layer so that the sender can send a lesser number of packets, so there is no flow control or congestion control mechanism. In this case, ICMP provides feedback, i.e., source quench. Suppose the sender resends the packet at a higher rate, and the router is not able to handle the high data rate. To overcome such a situation, the router sends a source quench message to tell the sender to send the packet at a lower rate.



The above diagram shows the message format of the source quench message. It is a type 4 message, and code is zero

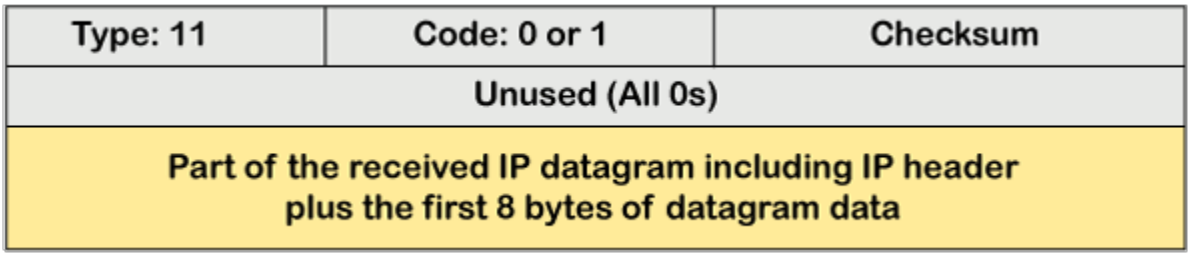
So, the sender must either stop or slow down the sending of datagrams until the congestion is reduced. The router sends one source-quench message for each datagram that is discarded due to the congestion in the network layer.

Time exceeded

Sometimes the situation arises when there are many routers that exist between the sender and the receiver. When the sender sends the packet, then it moves in a routing loop. The time exceeded is based on the time-to-live value. When the packet traverses through the router, then each router decreases the value of TTL by one. Whenever a router decreases a datagram with a time-to-live value to zero, then the router discards a datagram and sends the time exceeded message to the original source.

Each of the MAC layers has different data units. For example, some layers can handle upto 1500 data units, and some can handle upto 300 units. When the packet is sent from a layer having 1500 units to the layer having 300 units, then the packet is divided into fragments; this process is known as fragmentation. These 1500 units are divided into 5 fragments, i.e., f1, f2, f3, f4, f5, and these fragments reach the destination in a sequence. If all the fragments are not reached to the destination in a set time, they discard all the received fragments and send a time-exceeded message to the original source.

In the case of fragmentation, the code will be different as compared to TTL. Let's observe the message format of time exceeded.

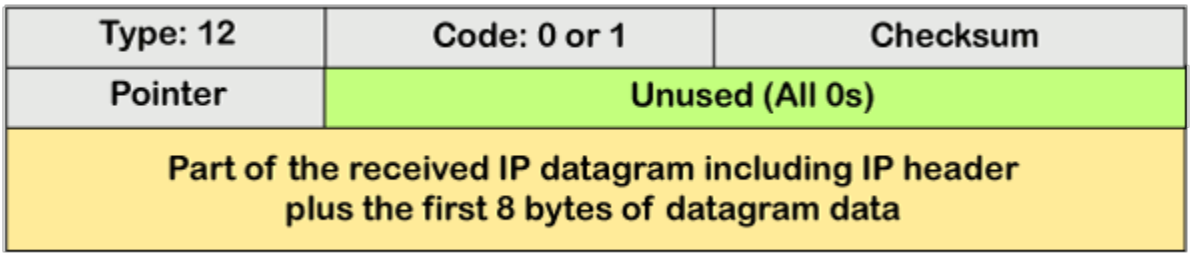


The above message format shows that the type of time-exceeded is 11, and the code can be either 0 or 1. The code 0 represents TTL, while code 1 represents fragmentation. In a time-exceeded message, the code 0 is used by the routers to show that the time-to-live value is reached to zero.

The code 1 is used by the destination to show that all the fragments do not reach within a set time.

Parameter problems

The router and the destination host can send a parameter problem message. This message conveys that some parameters are not properly set.

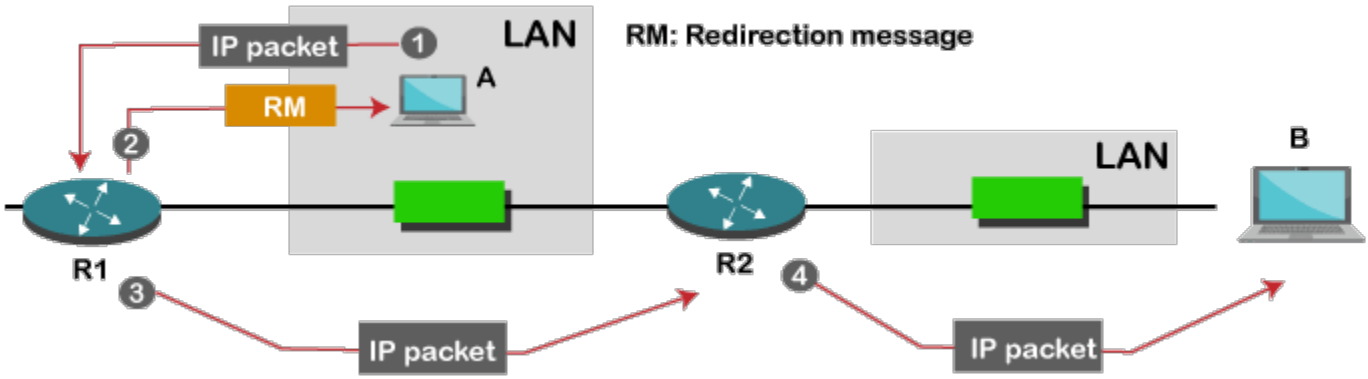


The above diagram shows the message format of the parameter problem. The type of message is 12, and the code can be 0 or 1.

Redirection

When the packet is sent, then the routing table is gradually augmented and updated. The tool used to achieve this is the redirection message. For example, A wants to send the packet to B, and there are two

routers exist between A and B. First, A sends the data to the router 1. The router 1 sends the IP packet to router 2 and redirection message to A so that A can update its routing table.



ICMP Query Messages

The ICMP Query message is used for error handling or debugging the internet. This message is commonly used to ping a message.

Echo-request and echo-reply message

A router or a host can send an echo-request message. It is used to ping a message to another host that "Are you alive". If the other host is alive, then it sends the echo-reply message. An echo-reply message is sent by the router or the host that receives an echo-request message.

Key points of Query messages

The echo-request message and echo-reply message can be used by the network managers to check the operation of the IP protocol. Suppose two hosts, i.e., A and B, exist, and A wants to communicate with host B. The A host can communicate to host B if the link is not broken between A and B, and B is still alive.The echo-request message and echo-reply message check the host's reachability, and it can be done by invoking the ping command.

The message format of echo-request and echo-reply message

Type 8: Echo request
Type 0: Echo reply

Type: 8 or 0	Code: 0	Checksum
Identifier		Sequence number
Optional data Sent by the request message; repeated by the reply message		

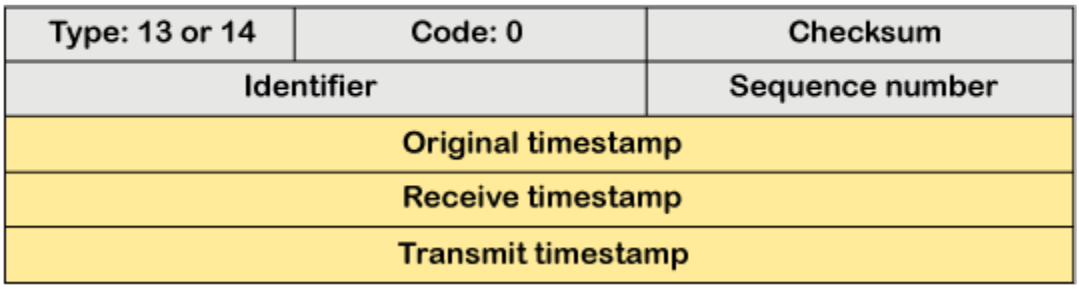
The above diagram shows the message format of the echo-request and echo-reply message. The type of echo-request is 8, and the request of echo-reply is 0. The code of this message is 0.

Timestamp-request and timestamp-reply message

The timestamp-request and timestamp-reply messages are also a type of query messages. Suppose the computer A wants to know the time on computer B, so it sends the timestamp-request message to computer B. The computer B responds with a timestamp-reply message.

Message format of timestamp-request and timestamp-reply

Type 13: request
Type 14: reply



The type of timestamp-request is 13, and the type of timestamp-reply is 14. The code of this type of message is 0.

Key points related to timestamp-request and timestamp-reply message

It can be used to calculate the round-trip time between the source and the destination, even if the clocks are not synchronized.

It can also be used to synchronize the clocks in two different machines if the exact transit time is known.

If the sender knows the exact transit time, then it can synchronize the clock. The sender asks the time on the receiver's clock, and then it adds the time and propagation delay. Suppose the time is 1:00 clock and propagation delay is 100 ms, then time would be 1:00 clock plus 100 ms.

Debugging tools

There are several tools used for debugging. In this topic, we will learn two tools that use ICMP for debugging. The two tools are ping and traceroute. We have learned about ping in echo-request and echo-reply messages that check whether the host or a router is alive or running.

Now we will take a look at the traceroute.

Traceroute is a tool that tracks the route taken by a packet on an IP network from source to destination. It records the time taken by the packet on each hop during its route from source to destination. Traceroute uses ICMP messages and TTL values. The TTL value is calculated; if the TTL value reaches zero, the packet gets discarded. Traceroute uses small TTL values as they get quickly expired. If the TTL value is 1 then the message is produced by router 1; if the TTL value is 2 then the message is produced by router 2, and so on.

Let's understand the traceroute through an example.

Suppose A and B are two different hosts, and A wants to send the packet to the host B. Between A and B, 3 routers exist. To determine the location of the routers, we use the traceroute tool.

TTL value =1: First, host A sends the packet to router 1 with TTL value 1, and when the packet reaches to router 1 then router reduces the value of TTL by one and TTL values becomes 0. In this case, router 1 generates the time-exceeded message and host A gets to know that router 1 is the first router in a path.

TTL value=2: When host A sends the packet to router 1 with TTL value 2, and when the packet reaches to router 1 then the TTL value gets decremented by 1 and the TTL value becomes 1. Then router 1 sends the packet to router 2, and the TTL value becomes 0, so the router generates a time-exceeded message. The host A gets to know that router 2 is the second router on the path.

TTL value=3: When host A sends the packet to router 1 with TTL value 3, then the router decrements its value by one, and the TTL value becomes 2. Then, router 1 sends the packet to router 2, and the TTL value becomes 1. Then, router 2 sends the packet to router 3, and the TTL value becomes 0. As TTL value becomes 0, router 3 generates a time-exceeded message. In this way, host A is the third router on a path.

RPL: (Routing Protocol for Low Power and Lossy Network)

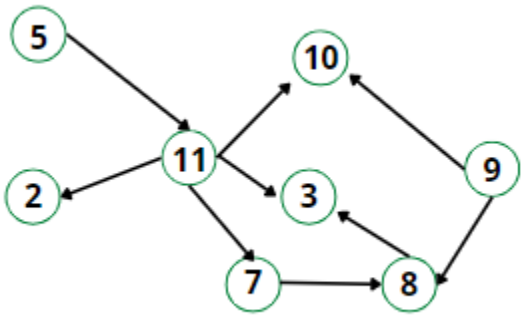
The term "RPL" stands for Routing Protocol for Low-Power and Lossy Networks. It is a specialized routing protocol designed specifically for use in Internet of Things (IoT) networks, where devices often have limited power resources, intermittent connectivity, and operate in environments with high packet loss.

RPL is an open standard protocol defined by the Internet Engineering Task Force (IETF). It is used to establish efficient routing paths between IoT devices in a network, enabling them to communicate with each other and with other parts of the network, such as gateway devices or cloud servers.

RPL stands for Routing Protocol for Low Power and Lossy Networks for heterogeneous traffic networks. It is a routing protocol for Wireless Networks. This protocol is based on the same standard as by Zigbee and 6 Lowpan is IEEE 802.15.4 It holds both many-to-one and one-to-one communication.

It is a Distance Vector Routing Protocol that creates a tree-like routing topology called the Destination Oriented Directed Acyclic Graph (DODAG), rooted towards one or more nodes called the root node or sink node.

The Directed Acyclic Graphs (DAGs) are created based on user-specified specific Objective Function (OF). The OF defines the method to find the best-optimized route among the number of sensor devices



Directed Acyclic Graph

The IETF chartered the ROLL (Routing Over Low power and Lossy networks) working group to evaluate all three routing protocols and determine the needs and requirements for developing a routing solution for IP smart objects. After the study of various use cases and a survey of existing protocols, the consensus was that a new routing protocol should be developed for IP smart objects, given the characteristics and requirements of the constrained network. This new Distance Vector Routing Protocol was named the IPv6 Routing Protocol for Low power and Lossy networks(RPL). The RPL specification was published as RFC 6550 by the ROLL working group.

In an RPL Network, each node acts as a router and becomes part of a mesh network. Routing is performed at the IP Layer. Each node examines every received IPv6 packet and determines the next-hop destination based on the information contained in the IPv6 header. No information from the MAC layer header is needed to perform the next determination.

Modes of RPL:

This protocol defines two modes:

1. Storing mode: All nodes contain the entire routing table of the RPL domain. Every node knows how to reach every other node directly.

2. Non-Storing mode: Only the border router(s) of the RPL domain contain(s) the full routing table. All other nodes in the domain maintain their list of parents only and use this as a list of default routes towards the border router. The abbreviated routing table saves memory space and CPU. When communicating in non-storing mode, a node always forwards its packet to the border router, which knows how to ultimately reach the final destination.

RPL is based on the concept of a Directed Acyclic Graph (DAG). A DAG is Directed Graph where no cycle exists. This means that from any vertex or point in the graph, we cannot follow an edge or a line back to

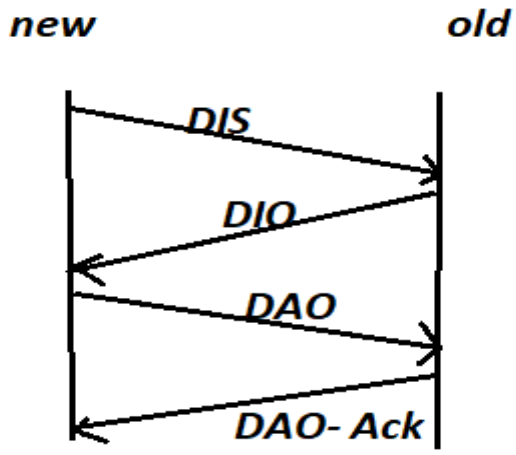
this same point. All of the edges are arranged in a path oriented toward and terminating at one or more root nodes.

A basic RPL process involves building a Destination Oriented Directed Acyclic Graph (DODAG). A DODAG is a DAG rooted in one destination. In RPL this destination occurs at a border router known as the DODAG root. In a DODAG, three parents maximum are maintained by each node that provides a path to the root. Typically one of these parents is the preferred parent, which means it is the preferred next hop for upward roots towards the root. The routing graph created by the set of DODAG parents across all nodes defines the full set of upwards roots. RPL protocol information should ensure that routes are loop-free by disallowing nodes from selected DODAG parents positioned further away from a border router.

Implementation of RPL Protocol:

The RPL protocol is implemented using the Contiki Operating system. This Operating System majorly focuses on IoT devices, more specifically Low Power Wireless IoT devices. It is an Open source Model and was first bought into the picture by Adam Dunkels.

The RPL protocol mostly occurs in wireless sensors and networks. Other similar Operating Systems include T-Kernel, EyeOS, LiteOS, etc.



Terminologies used in RPL:

- OF(Objective Function): its main purpose is to find the shortest path from all available path.
- Rank: it's the position of the node, from the root node.
- RPL Instance: if there are multiple RPL in a network ,every RPL is treated as one instance
- Root: it's always in Storing mode.
- Parent: parent is where arrow is pointing towards.
- Sub DODAG:it's a sub tree of given DODAG(Destination Oriented Directed Acyclic Graph)

- DODAG ID:-128 bits ID given by IPv6 to only root
- DODAG Version: Each new shape of tree is a version
- Grounded DODAG: DODAG which reached its goal is called grounded.
- Floating DODAG: DODAG which yet not reached its goal is called floating.

Control messages of RPL:

1) DIO:(DODAG Information Object)
   -This message is multicast downwards.
   -a given node in DODAG may multicast this message, which let other nodes know about it.
   -it announces other nodes" if they are interested to join?"

2) DIS(DODAG Information solicitation):

- When no announcement is heard and if a node wants to join DODAG, it sends a control message to know that if any DODAG exists

-message is like"is there any DODAG"

3) DAO(DODAG Advertisement Object):

-It's a request send by child to a parent or root.

-This message request allow the child to join DODAG.

4) DAO-Ack:

- DAO-Ack is a response sent by root.

- it can be yes or no

Key features of the RPL protocol include:

Energy efficiency: RPL is designed to minimize the energy consumption of IoT devices by optimizing routing paths. It takes into account the power constraints of individual devices and ensures that they participate in routing only when necessary.

Adaptive routing: RPL adapts to changes in the network topology and environmental conditions. It can dynamically reconfigure the routing paths based on factors such as device availability, link quality, and energy levels, ensuring reliable and efficient communication.

IPv6 compatibility: RPL is based on IPv6 addressing, allowing each IoT device to have a unique IP address. This facilitates end-to-end communication and integration with existing IP-based networks and internet infrastructure.

Parent-Child relationships: RPL organizes IoT devices in a hierarchical structure, with devices acting as parents and children to form a routing tree. This tree-like topology enables efficient routing and helps to conserve energy by reducing the number of hops between devices.

Objective Function (OF): RPL uses an Objective Function to evaluate and select the best routes based on specific metrics, such as link quality, available energy, or latency. The choice of the Objective Function depends on the requirements of the IoT application and can be customized accordingly.

RPL is widely used in IoT applications where devices need to communicate over low-power and lossy networks, such as smart grids, environmental monitoring systems, industrial automation, and home automation. It provides a reliable and energy-efficient solution for establishing communication paths in such challenging network environments.

CORPL

An extension of RPL is CORPL, or cognitive RPL, which is designed for cognitive networks and uses DODAG topology generation but with two new modifications to RPL. CORPL utilizes opportunistic forwarding to forward the packet by choosing multiple forwarders (forwarder set) and coordinates between the nodes to choose the best next hop to forward the packet to. DODAG is built in the same way as RPL.

Each node maintains forwarding set instead of its parent only and updates its neighbor with its changes using DIO messages. Based on the updated information, each node dynamically updates its neighbor priorities in order to construct the forwarder set.

"Cognitive RPL" (CRPL) is an extension of the RPL protocol in the context of IoT networks. It introduces cognitive capabilities to the routing decision-making process in order to enhance the overall performance and efficiency of the network.

CRPL leverages cognitive techniques, such as machine learning and artificial intelligence, to enable IoT devices to make intelligent routing decisions based on various factors and metrics. By analyzing network conditions, device capabilities, and application requirements, CRPL aims to optimize routing paths and adapt to dynamic changes in the network environment.

The cognitive aspects of CRPL include:

Learning and Adaptation: CRPL-enabled devices can gather and analyze data about network conditions, traffic patterns, and device capabilities. They can then use this information to learn and adapt their routing decisions over time. Machine learning algorithms can be applied to predict link quality, energy availability, or other relevant metrics to make informed routing choices.

Context-Awareness: CRPL takes into account contextual information, such as device location, network topology, and application requirements. By considering the context, CRPL-enabled devices can select routing paths that are most suitable for the specific situation, leading to improved performance and efficiency.

Dynamic Objective Functions: CRPL allows for the dynamic adjustment of objective functions used in the routing decision-making process. Objective functions define the metrics and goals for route selection. By dynamically adapting these functions based on changing conditions, CRPL can optimize routing paths based on real-time network dynamics.

The goal of Cognitive RPL is to enhance the efficiency, reliability, and adaptability of routing in IoT networks. By incorporating cognitive capabilities, CRPL aims to improve the overall performance of IoT applications and enable better utilization of network resources. However, it's important to note that the term "Cognitive RPL" may not have a standardized definition or implementation, as it could be specific to certain research or industry initiatives.

CARP:

Channel-Aware Routing Protocol (CARP) is a distributed routing protocol designed for underwater communication. It can be used for IoT due to its lightweight packets. It considers link quality, which is computed based on historical successful data transmission gathered from neighboring sensors, to select the forwarding nodes. There are two scenarios: network initialization and data forwarding.

In network initialization, a HELLO packet is broadcasted from the sink to all other nodes in the networks. In data forwarding, the packet is routed from sensor to sink in a hop- by-hop fashion. Each next hop is determined independently. The main problem with CARP is that it does not support reusability of previously collected data. In other words, if the application requires sensor data only when it changes significantly, then CARP data forwarding is not beneficial to that specific application.

An enhancement of CARP was done in E- CARP by allowing the sink node to save previously received sensory data. When new data is needed, E- CARP sends a Ping packet which is replied with the data from the sensors nodes. Thus, E-CARP reduces the communication overhead drastically

CARP, which stands for Context-Aware Routing Protocol, is a routing protocol designed for IoT (Internet of Things) networks. CARP aims to improve the efficiency and adaptability of routing decisions by considering context-aware information in the routing process.

In IoT environments, context-awareness refers to the ability of devices to gather and utilize contextual information about their surroundings. This information can include factors such as device location, network conditions, environmental data, device capabilities, and application requirements.

CARP utilizes this contextual information to make intelligent routing decisions. It considers factors such as energy consumption, link quality, device capabilities, and proximity to other devices or network resources. By taking into account the specific context of the IoT network, CARP can optimize routing paths and improve overall network performance.

The key features of CARP in IoT include:

Context Gathering: CARP-enabled devices gather contextual information from various sources, such as sensors, neighboring devices, or network infrastructure. This information can include location, signal strength, available energy, network congestion, and other relevant metrics.

Context Processing: CARP processes the gathered context information and analyzes it to determine the optimal routing decisions. Machine learning or rule-based algorithms can be employed to interpret the context and make informed routing choices.

Adaptive Routing: CARP adapts routing paths dynamically based on changes in the network context. It can react to variations in network conditions, device availability, or application requirements by selecting alternative routes or adjusting routing metrics.

Energy Efficiency: CARP considers energy consumption as a crucial factor in routing decisions. It aims to minimize energy usage by selecting routes that involve energy-efficient devices or by avoiding energy-draining paths.

Quality of Service (QoS) Support: CARP takes into account QoS requirements of IoT applications and can prioritize routing paths that meet specific QoS criteria, such as low latency, high reliability, or sufficient bandwidth.

CARP helps optimize routing decisions in IoT networks by leveraging context-aware information. By considering the specific characteristics and requirements of the IoT environment, CARP aims to enhance network performance, energy efficiency, and overall user experience.

Features	RPL	CORPL	CARP
Full Form	Routing Protocol for Low-Power and Lossy Networks	Cognitive RPL	Channel-Aware Routing Protocol
Server technologies	Supported	Supported	Not supported
Security	Not supported	Not supported	Not supported
Storage management	Supported	Not supported	Supported
Data management	Supported	Supported	Supported